

AIM - Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

b) Convert wide data into tall format

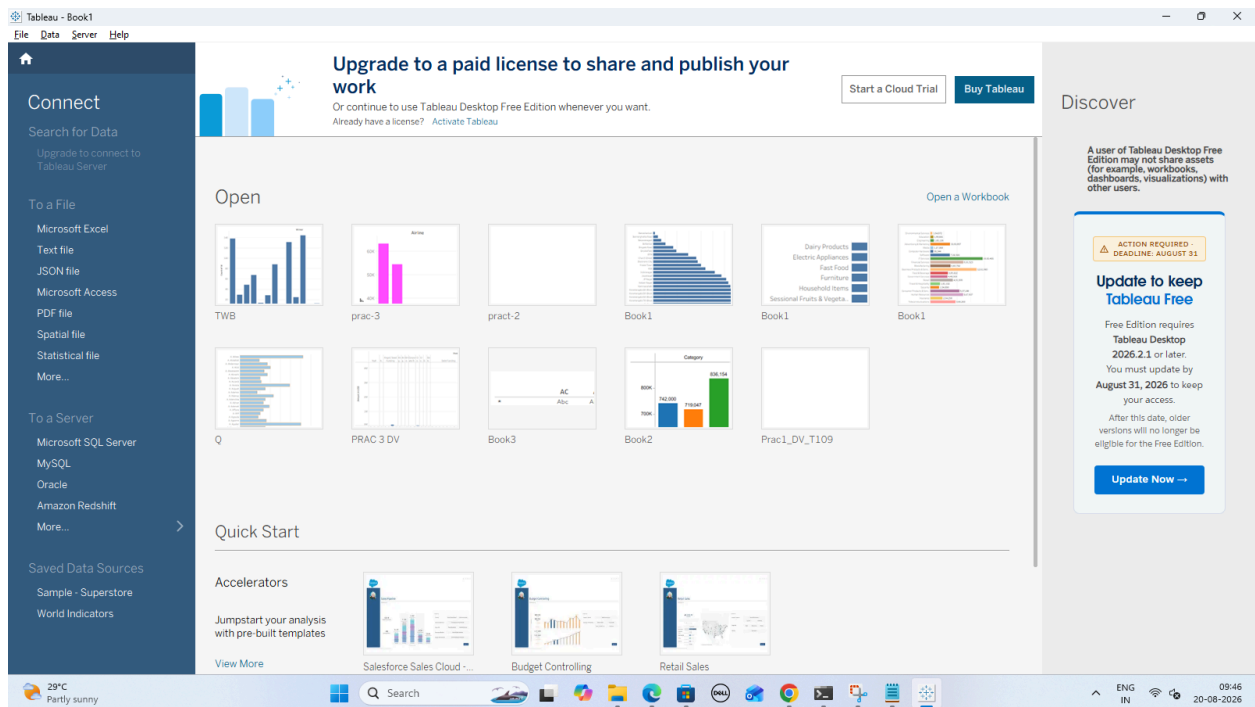
c) Perform union of multiple files

d) Execute cross-database joins

e) Restructure data for better visualization

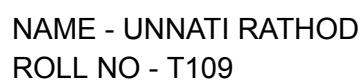
f) Handle different levels of detail in datasets

g) Apply advanced data cleaning techniques



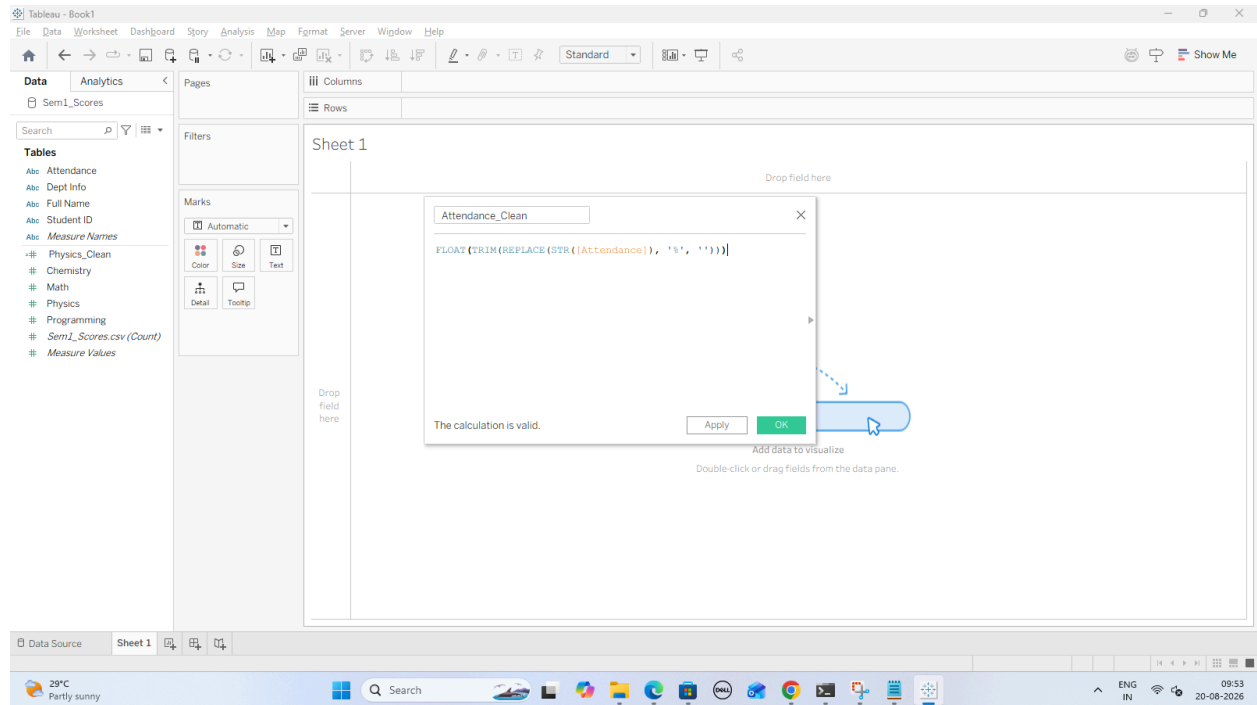
A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

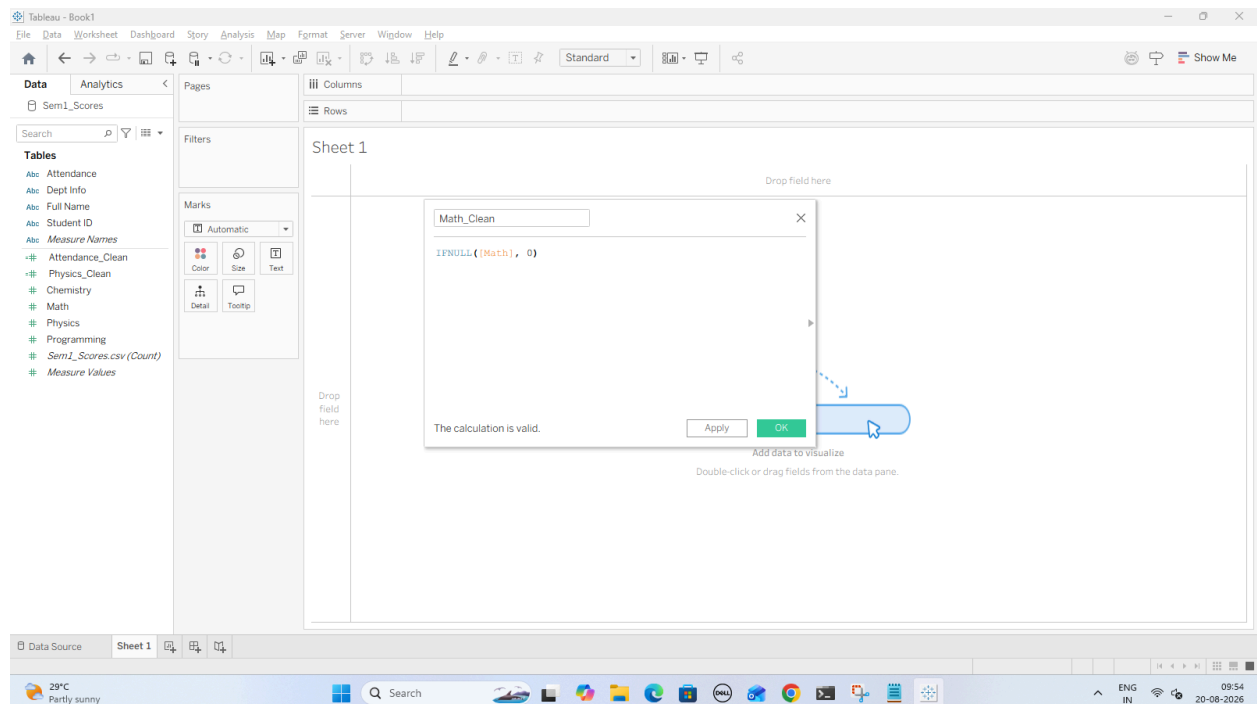


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Step 3: Handle Missing (Null) Numeric Values via Imputation



Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)



B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

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ROLL NO - T109

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores.csv

12 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
#	Math	Sem1_Scores.csv	Math

Abc	Abc	Abc	#	#	#	#	Abc
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

Step 2: Multi-Select and Pivot Columns

Tableau - Book1

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Sem1_Scores.csv

12 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote Field
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance

Abc	Abc	Abc	#	#	#	#	Abc
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %

Step 3: Rename the Pivoted Fields

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Tableau - Book1

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Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores.csv

11 fields 30 rows

30 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Abc	Attendance	Sem1_Scores.csv	Attendance
Abc	Subject	Pivot	Pivot Field Na...
#	Score	Pivot	Pivot Field Val...
Abc	Physical Info	Calculation	Calculation L...

Student ID	Full Name	Dept Info	Math	Attendance	Subject	Pivot Field Values
S101	Aarav Sharma	CSE-01	85	85 %	Chemistry	88.0000
S101	Aarav Sharma	CSE-01	85	85 %	Physics	0.9000
S101	Aarav Sharma	CSE-01	85	85 %	Programming	95.0000
S102	priya VERMA	ECE-02	null	90 %	Chemistry	80.0000
S102	priya VERMA	ECE-02	null	90 %	Physics	0.7500
S102	priya VERMA	ECE-02	null	90 %	Programming	85.0000
S103	Rohan Gupta	CSE-01	78	78 %	Chemistry	null

Step 4: Clean and Format the Tall

Tableau - Book1

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Text file

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Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores.csv

11 fields 30 rows

30 rows

Name Sem1_Scores.csv

Description
No description available.

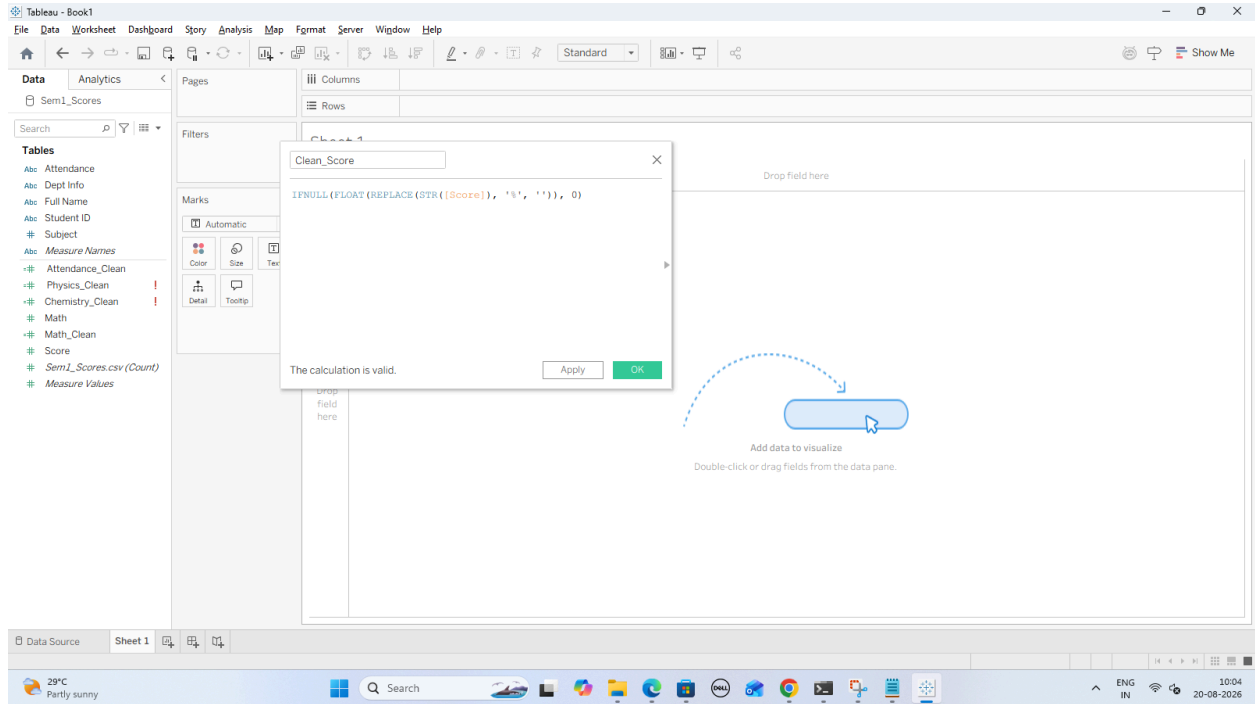
Fields

Type	Field Name	Physical Table	Remote Field
Abc	Attendance	Sem1_Scores.csv	Attendance
Abc	Subject	Pivot	Pivot Field Na...
#	Score	Pivot	Pivot Field Val...
Abc	Physical Info	Calculation	Calculation L...

Student ID	Full Name	Dept Info	Math	Attendance	Subject	Score
S101	Aarav Sharma	CSE-01	85	85 %	null	88.0000
S101	Aarav Sharma	CSE-01	85	85 %	null	0.9000
S101	Aarav Sharma	CSE-01	85	85 %	null	95.0000
S102	priya VERMA	ECE-02	null	90 %	null	80.0000
S102	priya VERMA	ECE-02	null	90 %	null	0.7500
S102	priya VERMA	ECE-02	null	90 %	null	85.0000
S103	Rohan Gupta	CSE-01	78	78 %	null	null

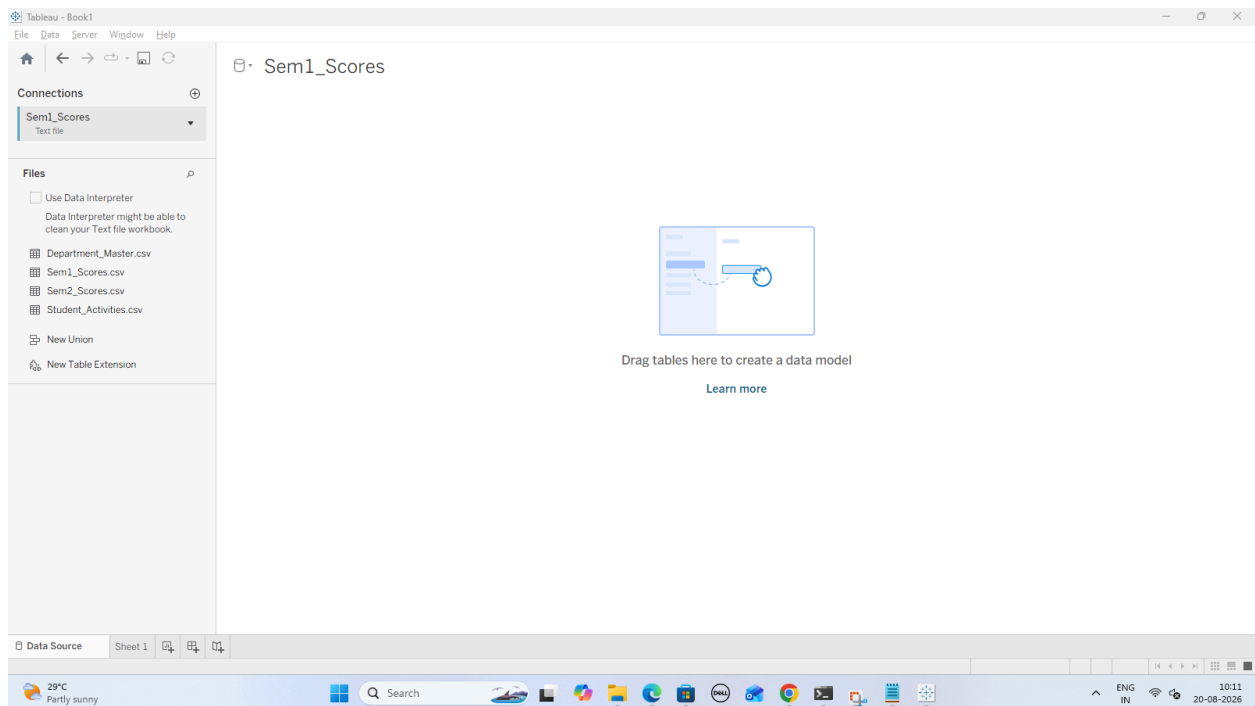
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C: Perform Union of Multiple Files

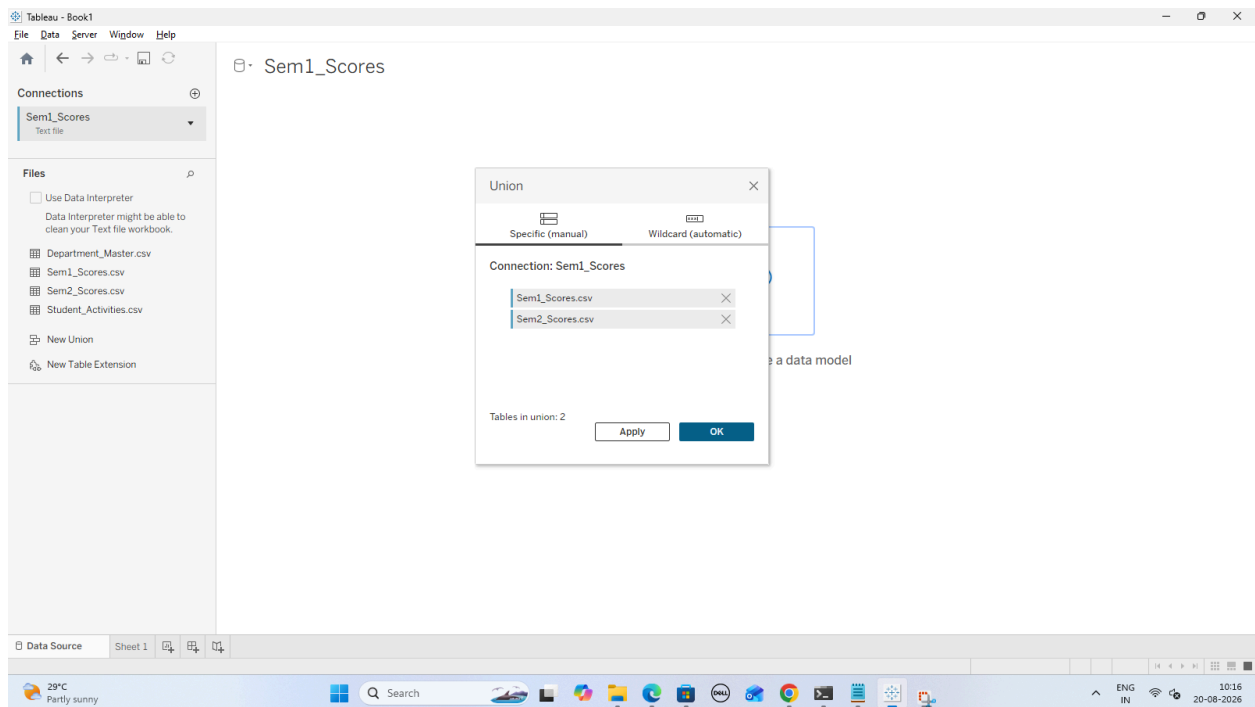
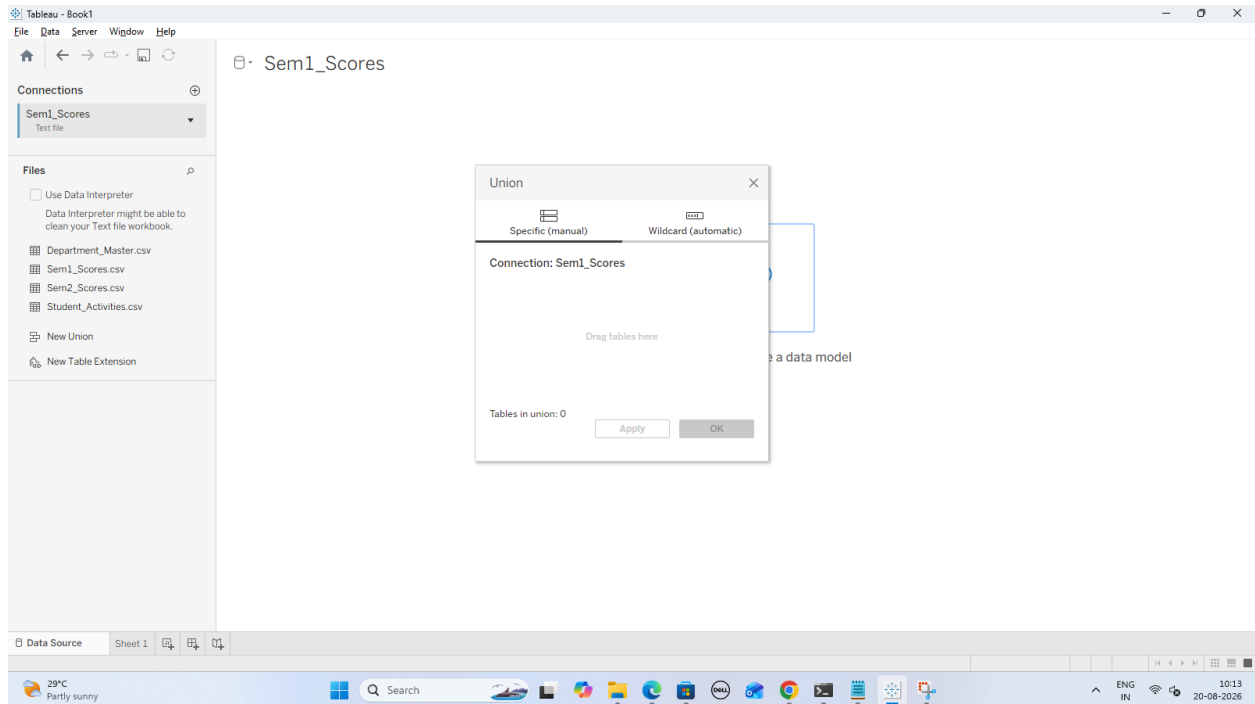
Step 1: Open the Data Source Canvas



Step 2: Create a Multi-File Union

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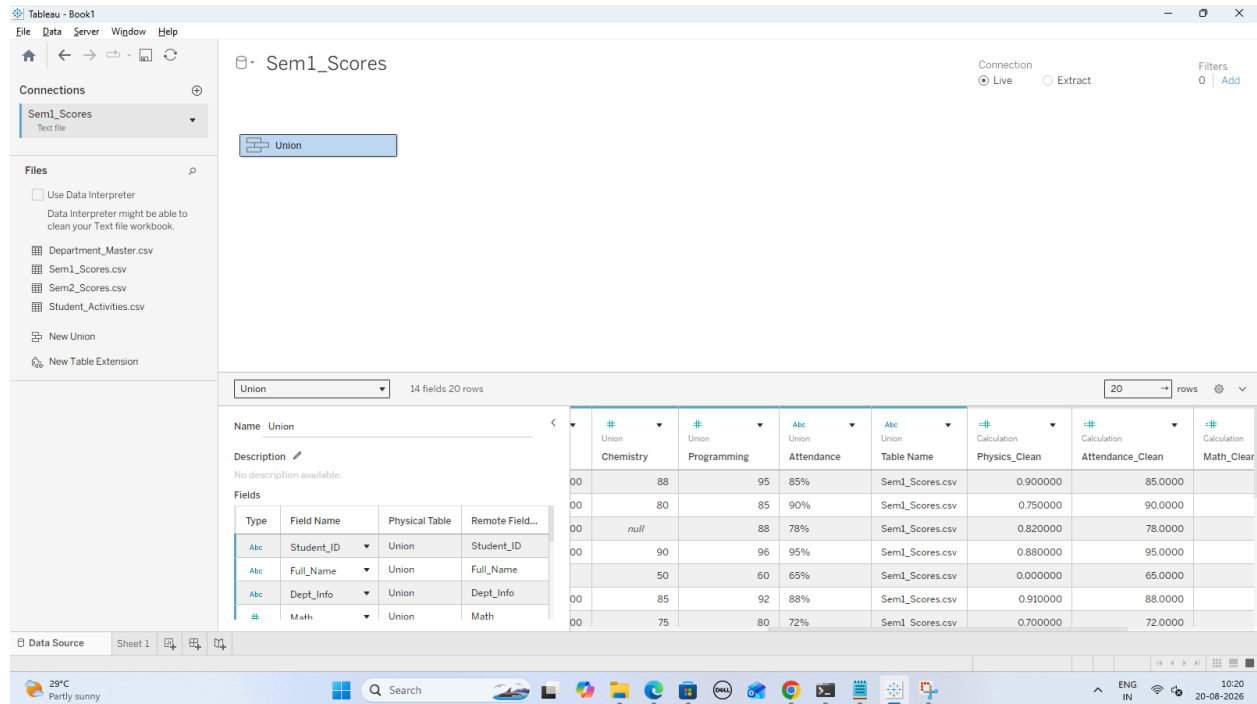
Step 3: Verify the Merged Data Grid

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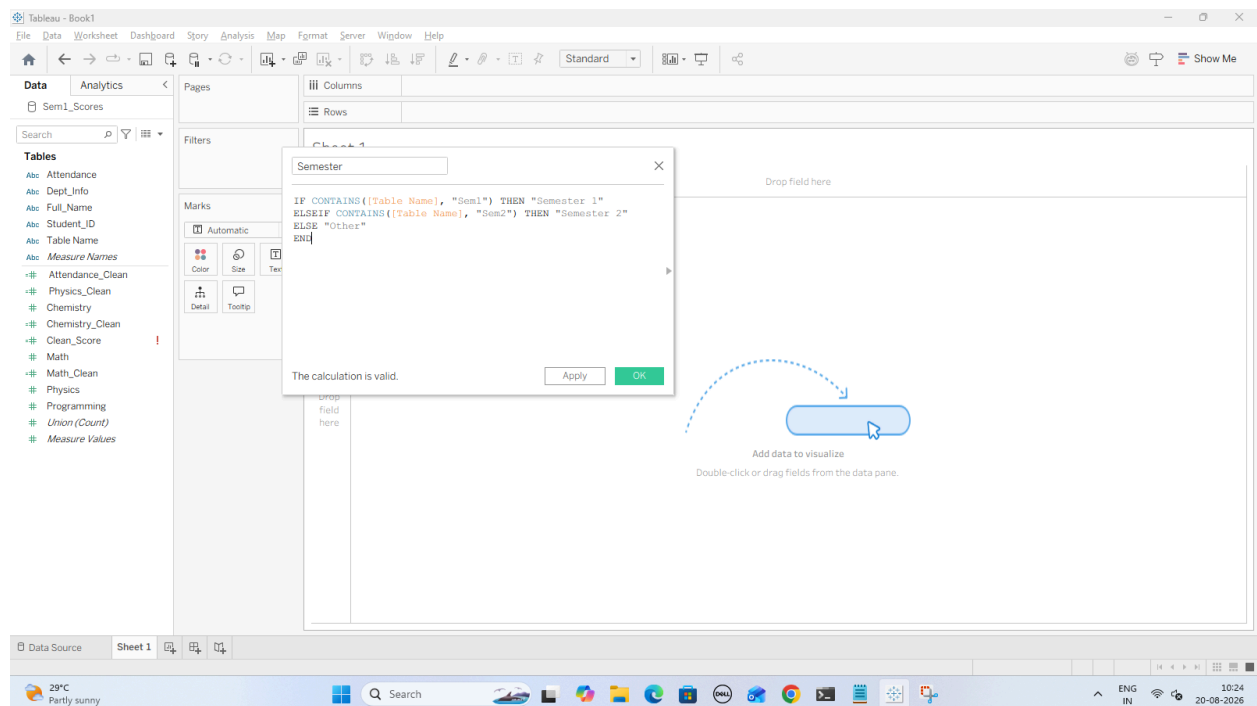
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Step 4: Create a Clean Semester Identifier Dimension



- . D: Execute Cross-Database Joins
- Step 1: Stay on the Top Canvas
- Step 2: Add and Link Department_Master.csv

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract Filters: 0 Add

Union — Department...

4 rows

Name: Department_Master.csv
How do relationships differ from joins? Learn more

Union Operator Department_Maste...

Abc Dept_Info = Abc Dept Code

Add more fields

Performance Options

Remove Relationship

Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Step 3: Add and Link Student_Activities.csv

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract Filters: 0 Add

Union — Student_A...

10 rows

Name: Student_Activities.csv
How do relationships differ from joins? Learn more

Union Operator Student_Activities...

Abc Student_ID = Abc Student ID (Stu...

Add more fields

Performance Options

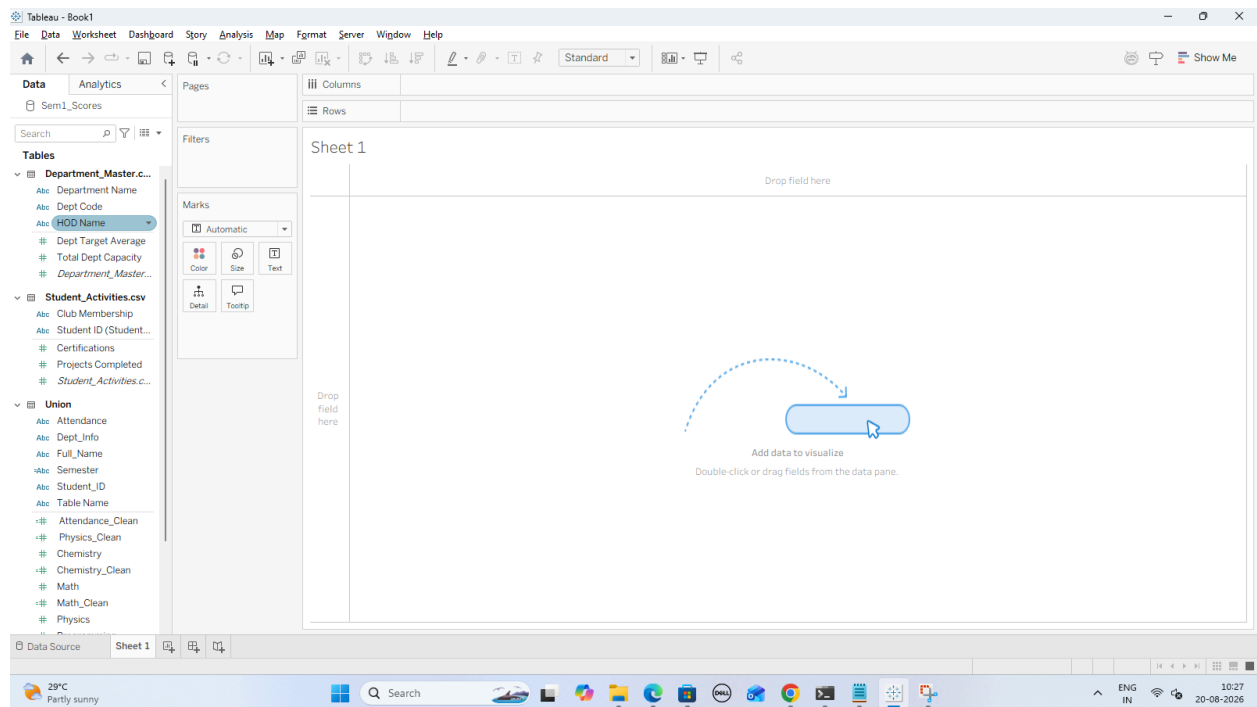
Remove Relationship

Student ID (Student Activ)	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1

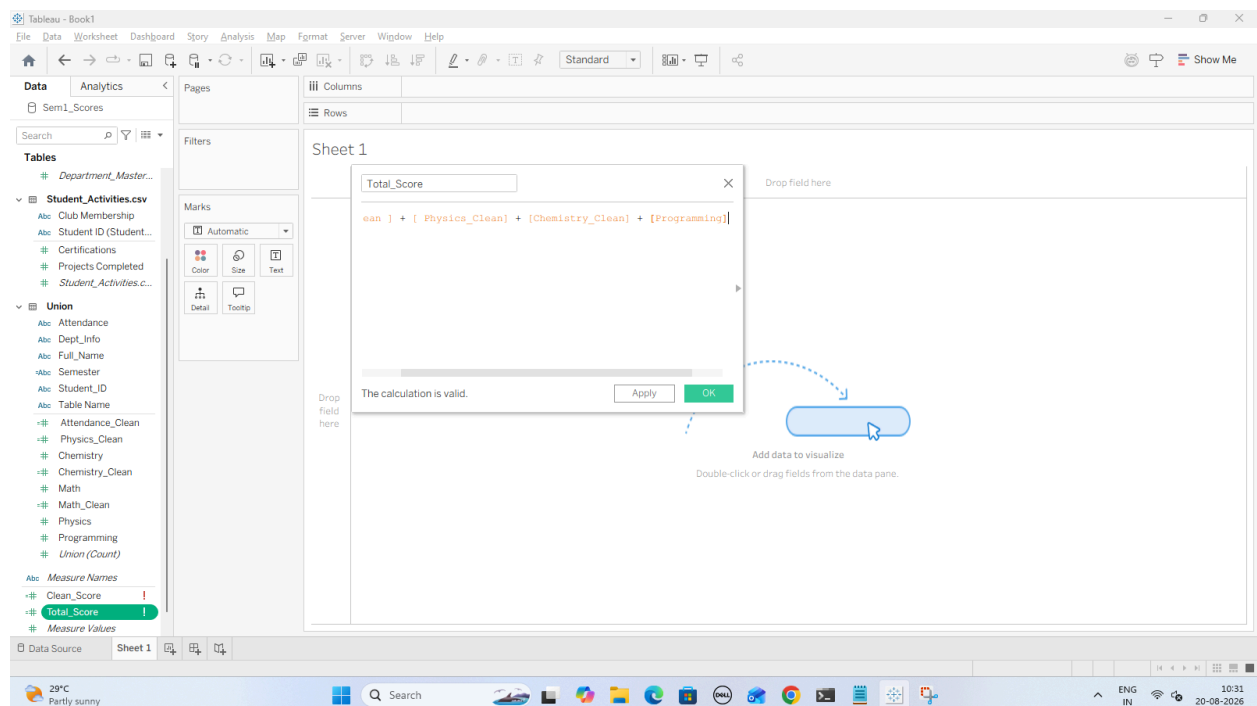
Step 4: Verify in Sheet 1

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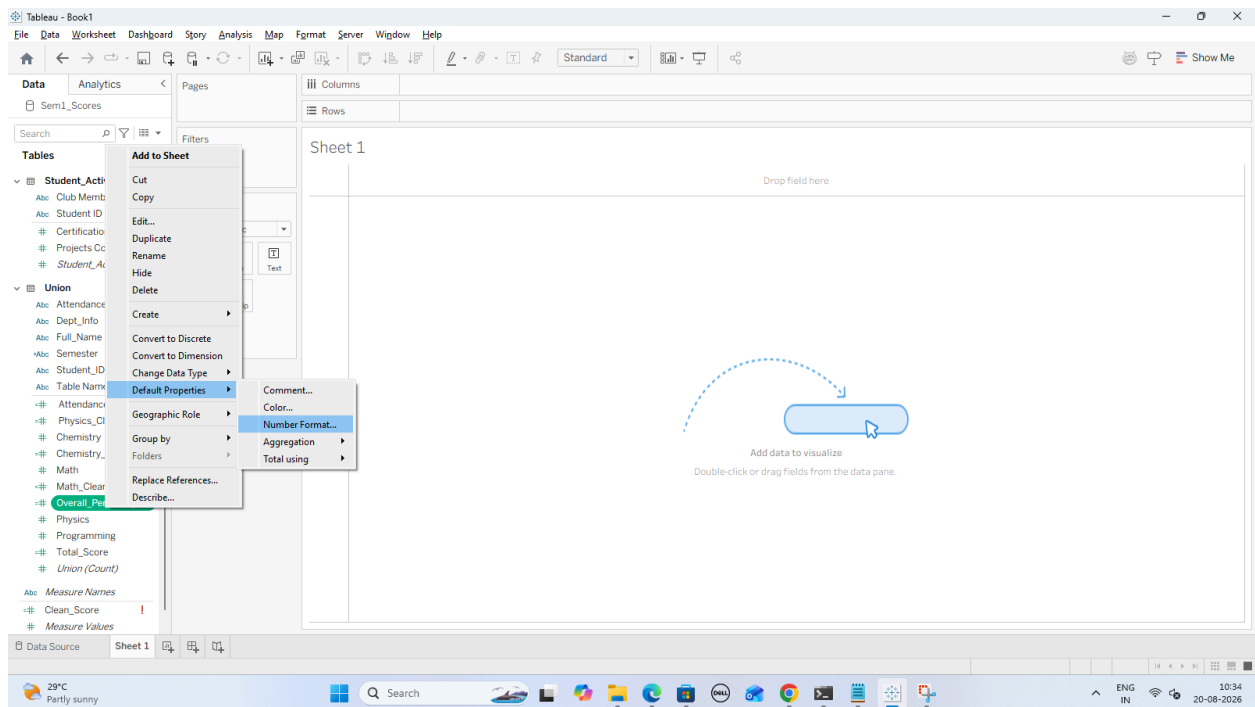
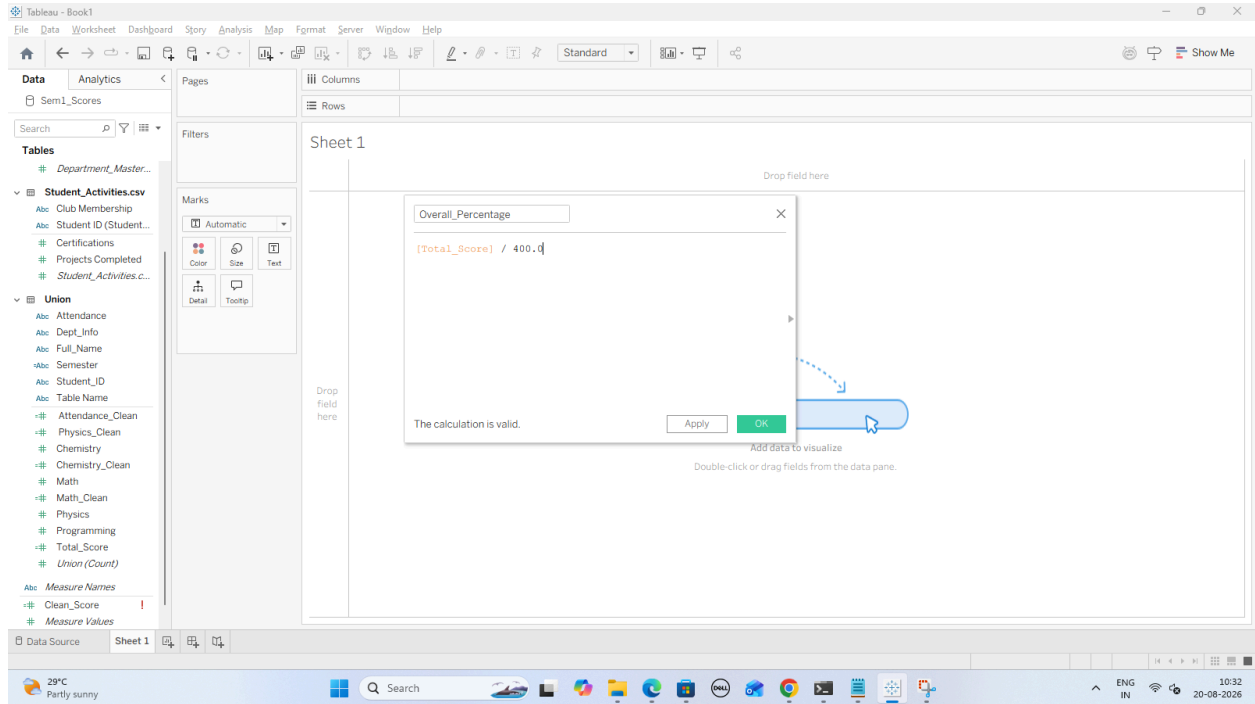


E: Restructure Data for Better Visualization
Step 1: Calculate Total Marks and Scaled Percentage



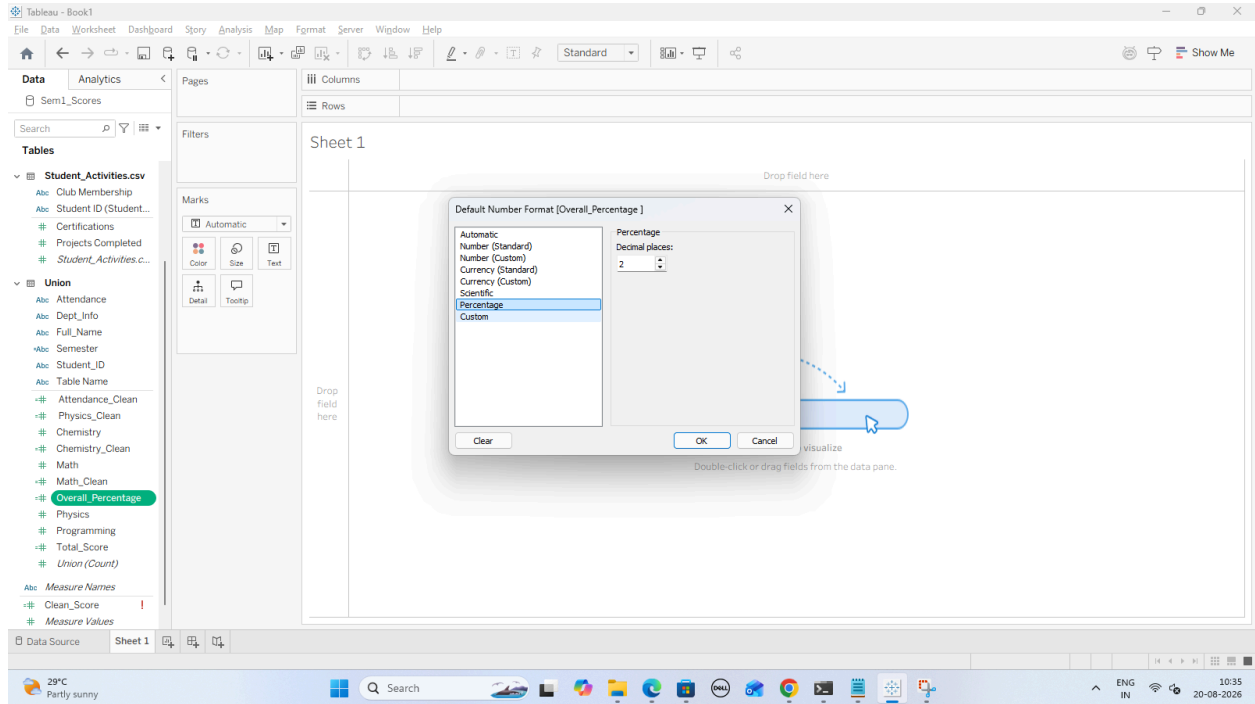
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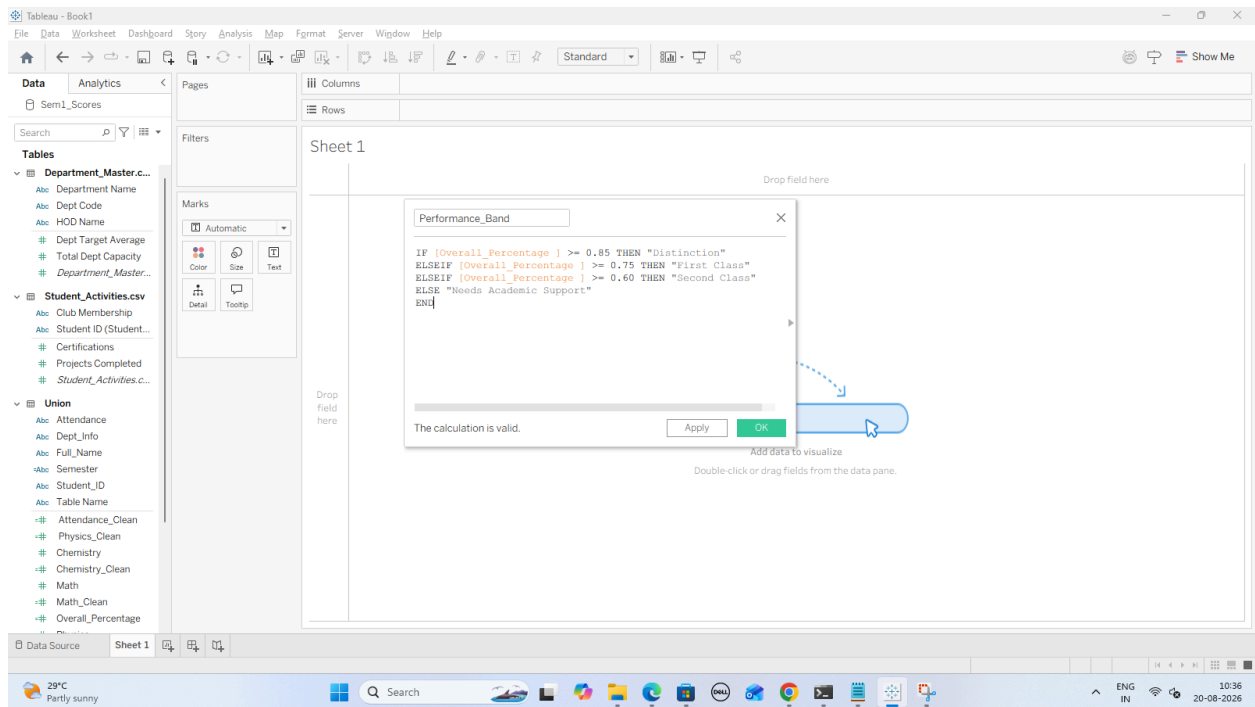


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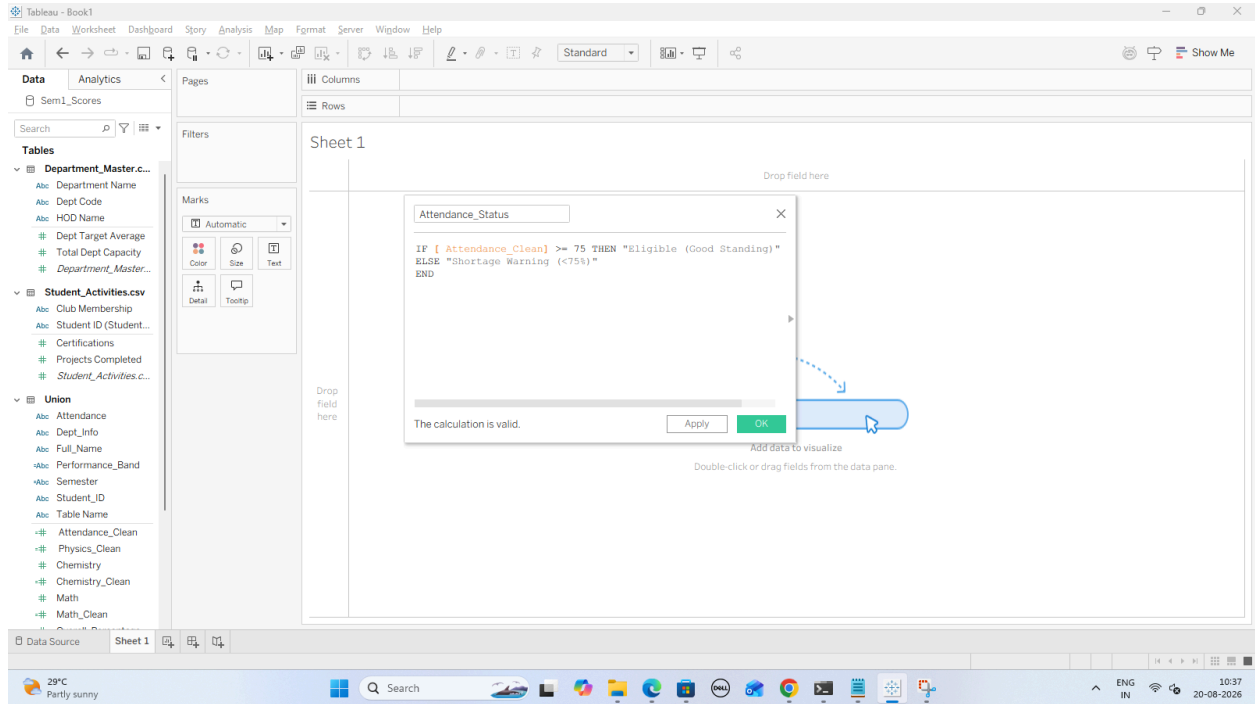
Step 2: Create Categorical Performance Bands (Grading Tiers)



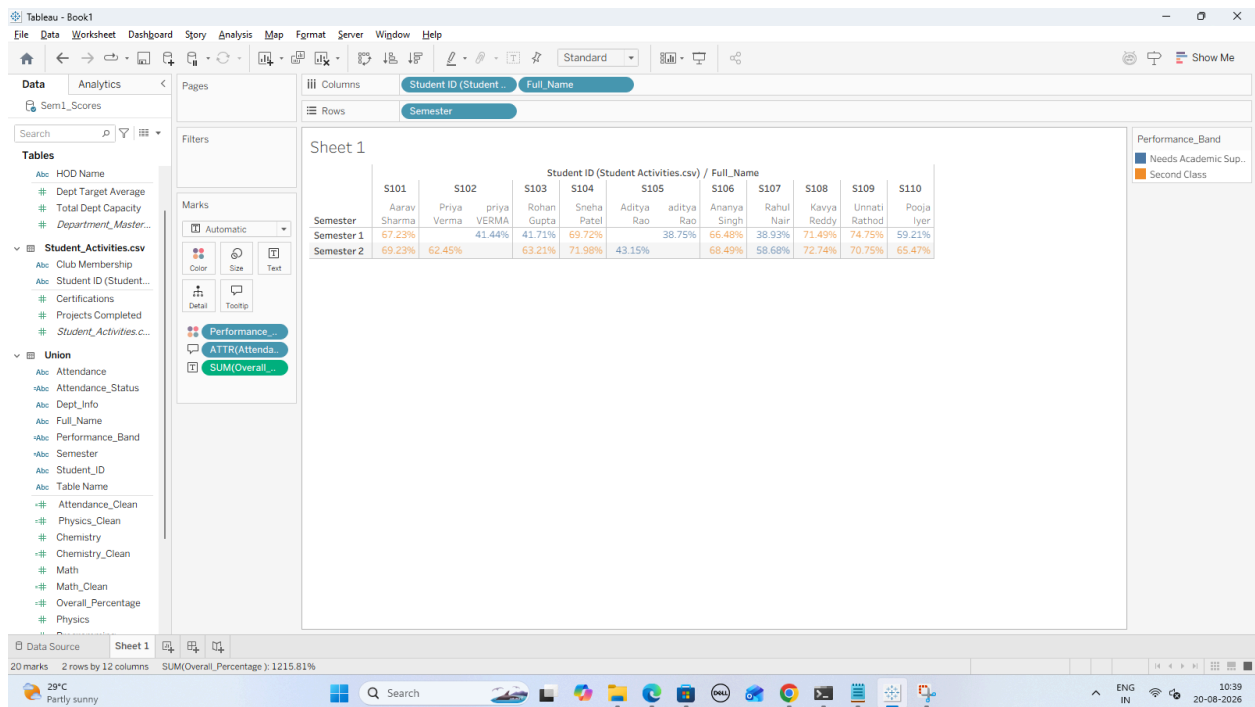
Step 3: Create an Attendance Health Flag

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Step 4: Build the Student Performance Scorecard View

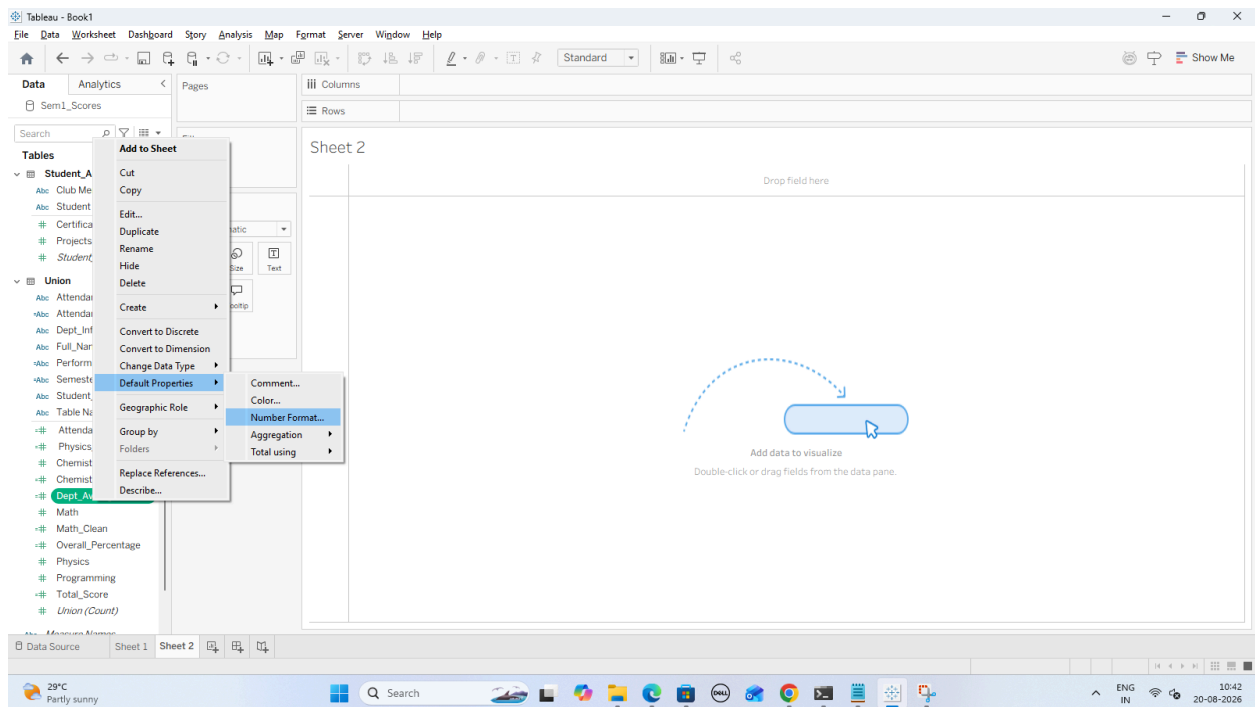
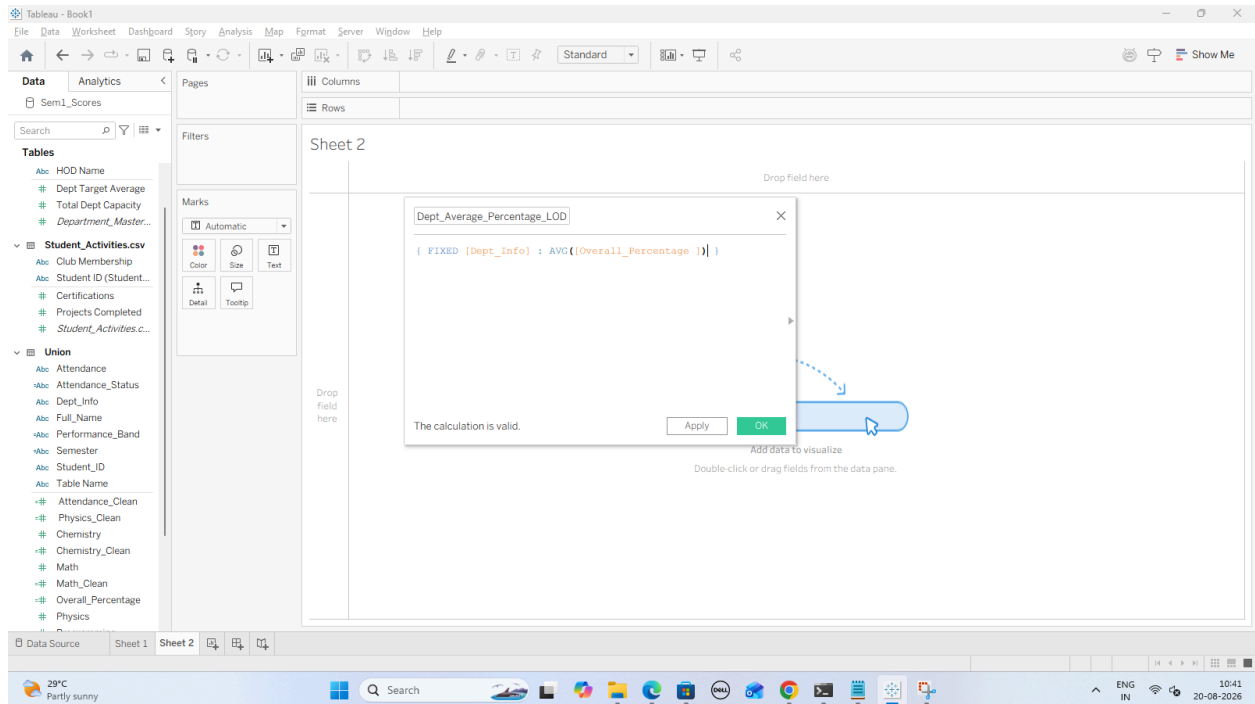


F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

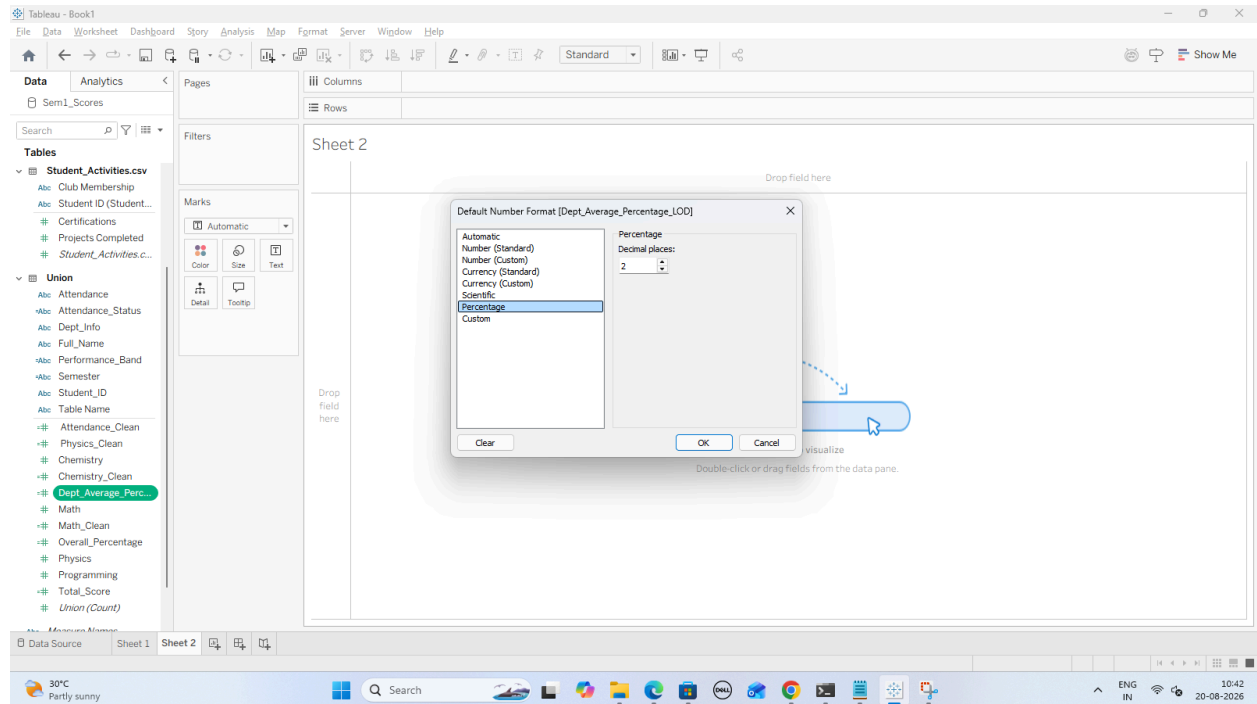
NAME - UNNATI RATHOD
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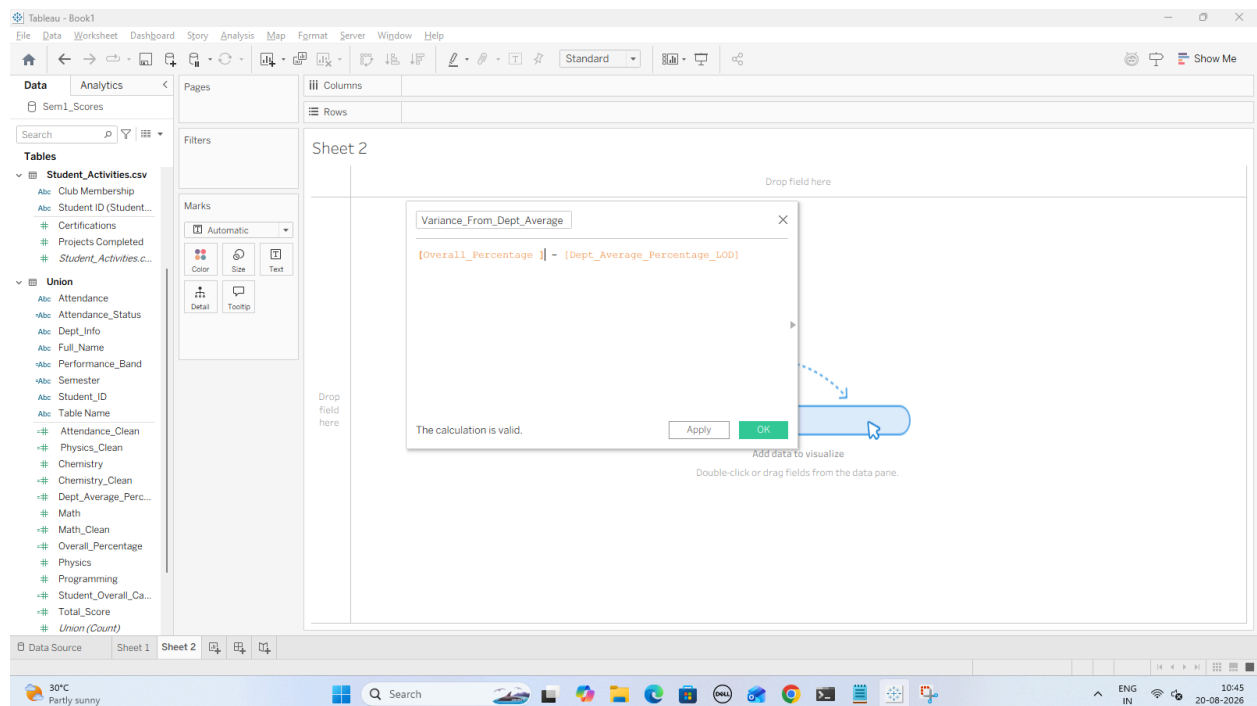


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Step 2: Compute Student Multi-Semester Average using {FIXED} LOD



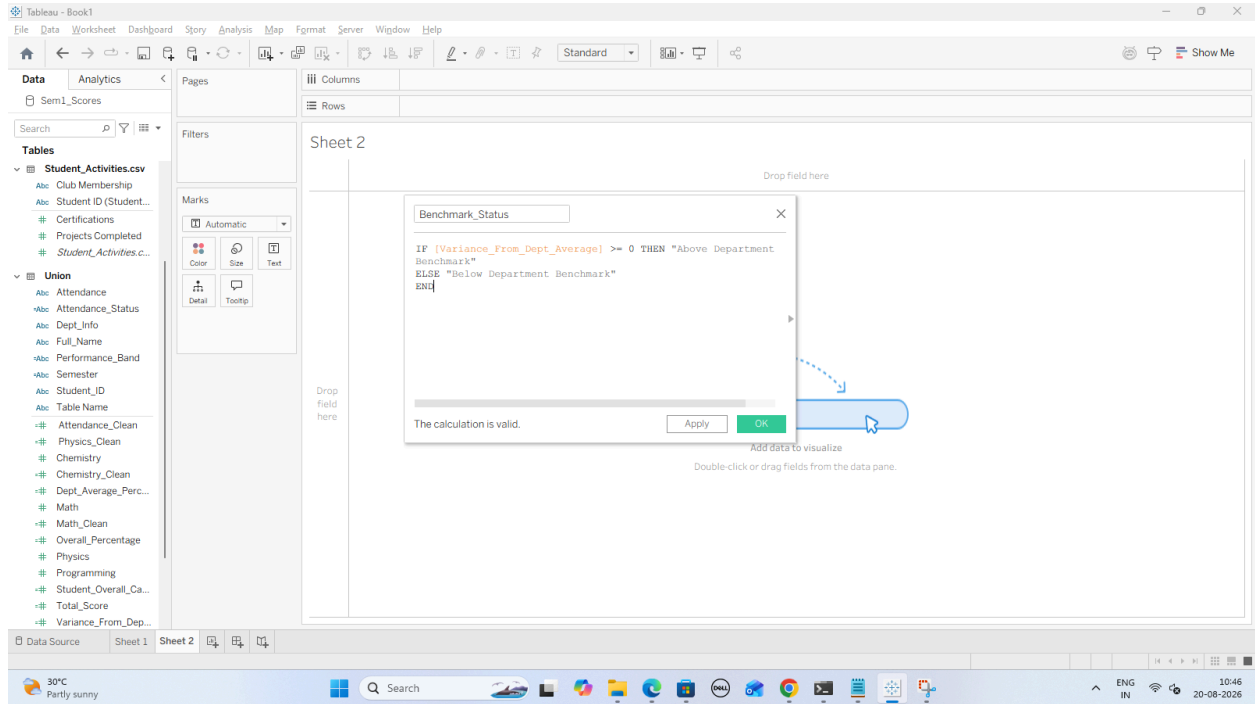
Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

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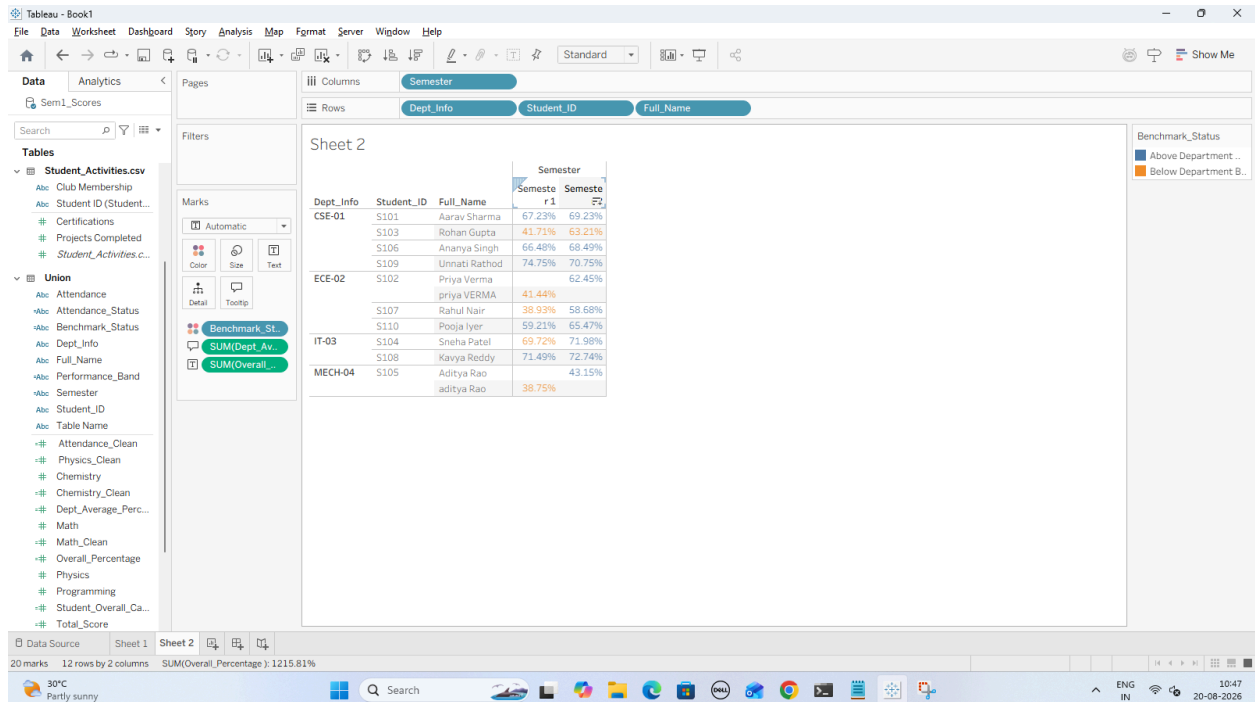
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Step 4: Create a Benchmark Indicator Dimension

Step 5: Build the LOD Benchmark Visualization



G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

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The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Student ID (Student Activities.csv)' and 'Full_Name'. The 'Rows' shelf contains 'Semester'. A calculation dialog box is open for the field 'Full_Name_Clean'. The calculation formula is: `UPPER(LEFT(TRIM({Full_Name}), 1)) + LOWER(MID(TRIM({Full_Name}), 2, FIND(TRIM({Full_Name}), " ") - 1)) + " " + UPPER(MID(TRIM({Full_Name}), FIND(TRIM({Full_Name}), " ") + 1)) + LOWER(MID(TRIM({Full_Name}), FIND(TRIM({Full_Name}), " ") + 1, LEN(TRIM({Full_Name}))))`. The dialog also shows a preview of the data table with columns S101 through S110 and rows for Semester 1 and Semester 2. The status bar at the bottom indicates '20 marks' and '2 rows by 12 columns'.

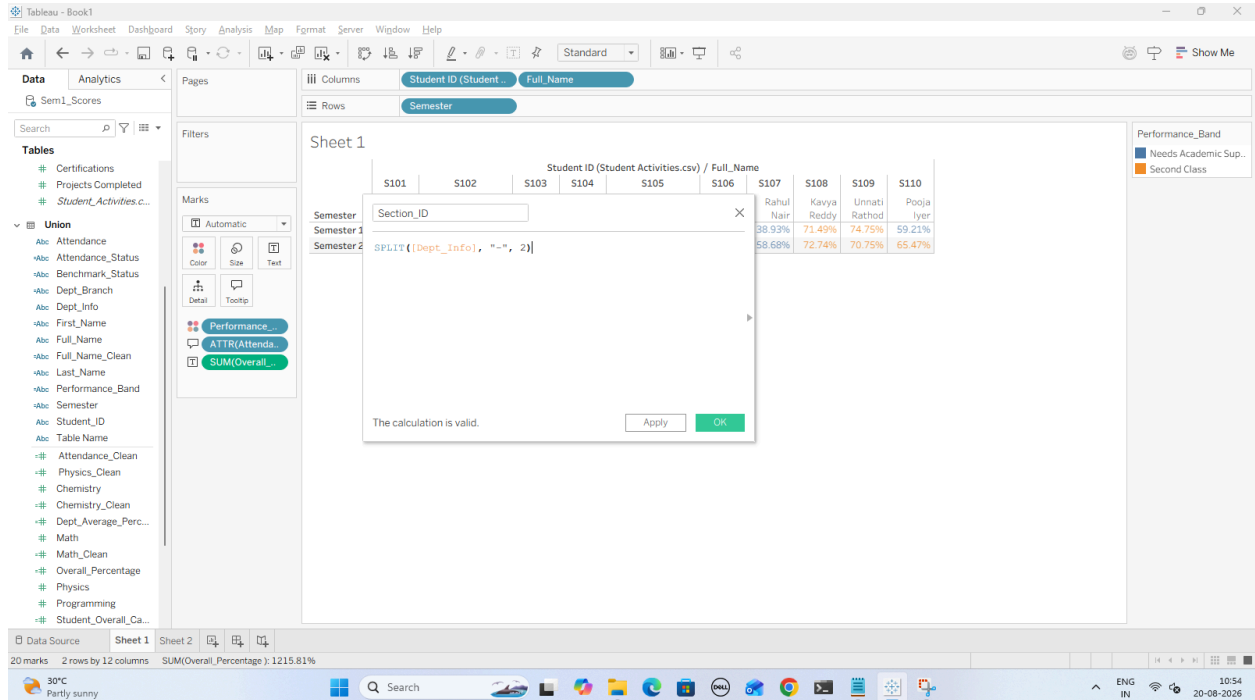
Step 2: Parse First Name and Last Name using SPLIT()

The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Student ID (Student Activities.csv)' and 'Full_Name'. The 'Rows' shelf contains 'Semester'. A calculation dialog box is open for the field 'First_Name'. The calculation formula is: `TRIM(SPLIT({Full_Name}, " ", 1))`. The dialog also shows a preview of the data table with columns S101 through S110 and rows for Semester 1 and Semester 2. The status bar at the bottom indicates '20 marks' and '2 rows by 12 columns'.

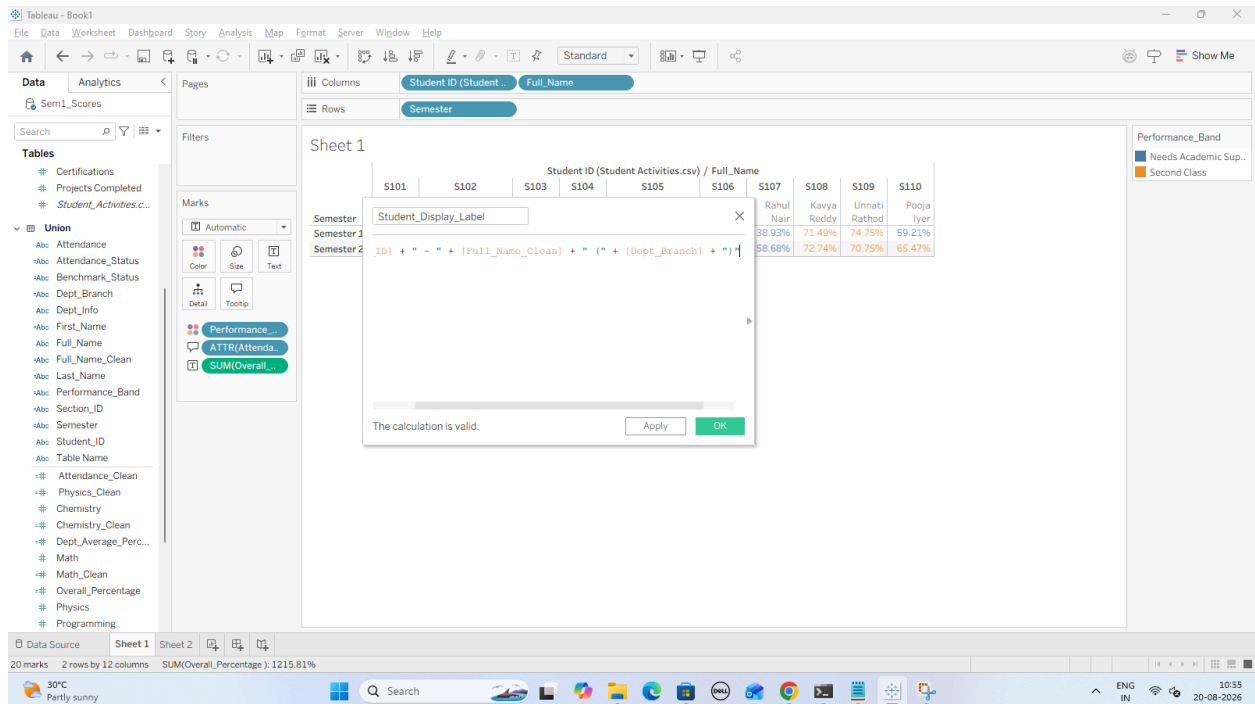
Step 3: Split Composite Department Codes (Dept_Info)

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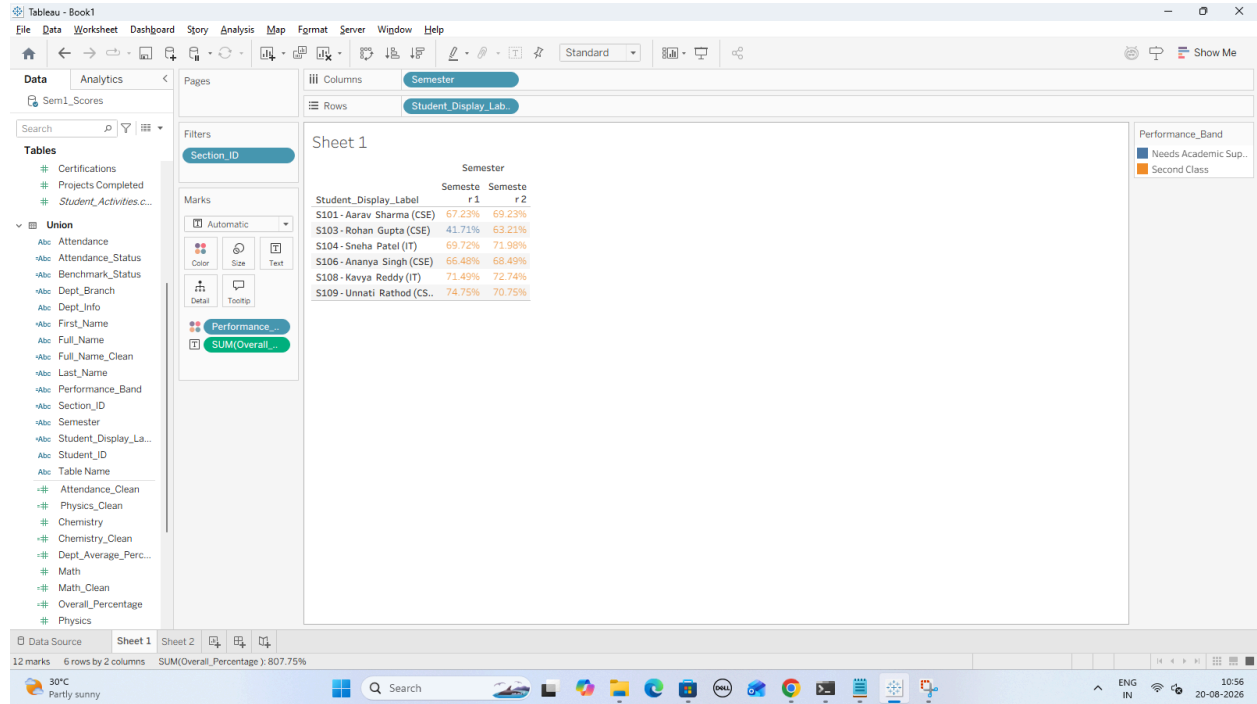
Step 4: Create a Clean Composite Student Label



Step 5: Verify in the Final Dashboard View

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